

Basic Interface Implementation in Java

Basic Information

Assignment: Assignment 2

Program: BasicInterfaceDemo.java

Topic: Interfaces — Different Types of Implementations of Interfaces

Objective / Aim

To demonstrate the basic implementation of an interface in Java by declaring a method in an interface and implementing it in a class.

System Requirements

Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the Animal interface with the sound() method.
3. Create the Dog class and implement the Animal interface.
4. Define sound() in Dog to display the message "Dog barks".
5. Create a Dog object in main().
6. Call the sound() method using the Dog object.
7. Stop the program.

UML Class Diagram



Source Code

```
// Interface
interface Animal {

void sound();
}

// Class implements the interface
class Dog implements Animal {

public void sound() {
System.out.println("Dog barks");
}
}

public class BasicInterfaceDemo {
public static void main(String[] args) {

// Create object of implementing class
Dog dog = new Dog();

dog.sound();
}
}
```

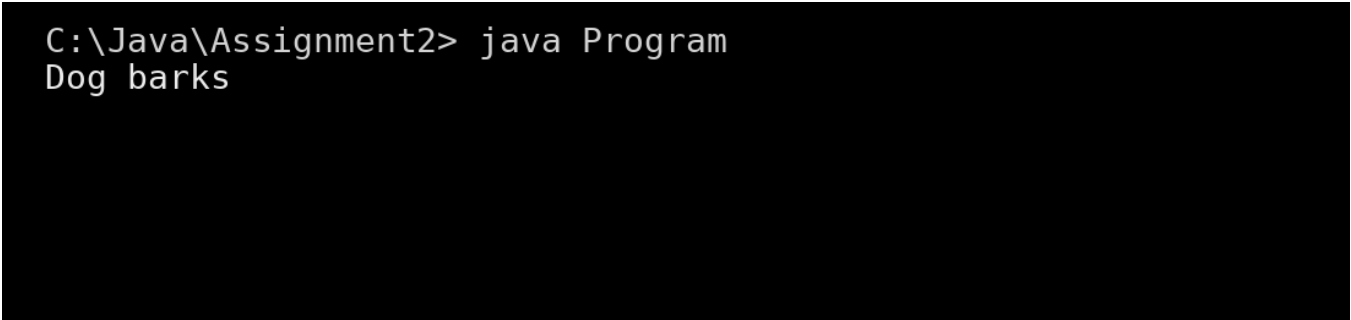
Compilation & Execution Commands

Save as: BasicInterfaceDemo.java

Compile: javac BasicInterfaceDemo.java

Execute: java BasicInterfaceDemo

Output Screen



```
C:\Java\Assignment2> java Program
Dog barks
```

Result

Thus, the basic interface implementation program was implemented successfully using the interface concepts demonstrated in the source code.